

A Discriminant Barrier for High- ℓ Kummer Non-Concentration and a Bilinear Equivalence for the Erdős–Graham #203 Seam

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Abstract

The Erdős–Graham #203 program reduces, conditionally, to a uniform prime Hecke non-concentration (PHNC) statement in the high- ℓ seam $(\log Q)^{1-\varepsilon} < \ell \leq (\log Q)^B$. In this note we establish two structural theorems that sharpen the conditional landscape.

The first (the *Discriminant Barrier*) is a quantitative ceiling: any bound on the bad-character set $|B_\eta(\ell, Q)|$ obtained from a log-free zero-density estimate applied to the rank-two Kummer Hecke family over $\mathbb{Q}(\zeta_\ell)$, using only the analytic conductor $\mathfrak{q}(\chi) \cdot |d_{K_\ell}|$, is capped at $\ell \ll \log Q / \log \log Q$. The intrinsic obstruction is the cyclotomic discriminant $|d_{K_\ell}| = \ell^{\ell-2}$ entering the analytic conductor; crossing it requires a degree-uniform Deuring–Heilbronn theorem over the cyclotomic tower $\{K_\ell\}$, which is not in the literature.

The second is a *Bilinear-Kummer Equivalence*: PHNC at level ℓ is logically equivalent (modulo unconditionally provable Type-I bounds via the exact Γ -fiber identity) to a precisely stated Type-II Kummer Bilinear estimate TKB_ℓ uniformly over hyperplane modes $A \in \mathbb{F}_\ell^r \setminus 0$. We locate TKB_ℓ in the analytic hierarchy: it is strictly stronger than the Heath–Brown cubic large sieve and strictly weaker than the relevant family GRH. As a consequence, the Friedlander–Iwaniec asymptotic sieve for primes does not deliver PHNC even granting TKB_ℓ because the Erdős–Graham orbit equidistributes against Kummer- Γ local densities $|\Gamma_q|/(q-1)$ rather than the uniform densities $1/\varphi(q)$; a Kummer-twisted asymptotic sieve framework is required.

This note does not claim a resolution of Erdős–Graham Problem #203. Its purpose is to make the conditional analytic seam precise enough to commit the next round to a specific named bilinear estimate and a specific cyclotomic-tower Deuring–Heilbronn target.

1 Background and notation

We work in the setup of the companion papers [8, 9, 12]. Fix an odd prime ℓ and let

$$K_\ell = \mathbb{Q}(\zeta_\ell), \quad K_{\ell,\star} = K_\ell(2^{1/\ell}, 3^{1/\ell}), \quad [K_\ell : \mathbb{Q}] = \ell - 1, \quad [K_{\ell,\star} : K_\ell] = \ell^2,$$

where the second equality is the rank-two Kummer degree theorem (Paper 6, Corollary on rank- r star field degree). For $A = (A_1, A_2) \in \mathbb{F}_\ell^2 \setminus \{0\}$, define

$$\alpha_A = 2^{A_1} 3^{A_2} \in \mathbb{Q}^*, \quad K_{\ell,A} = K_\ell(\alpha_A^{1/\ell}).$$

There are $\ell + 1$ projective lines in $\mathbb{F}_\ell^2$, hence $\ell + 1$ one-radical Kummer subextensions $K_{\ell,A} \subset K_{\ell,\star}$, and the hyperplane Kummer fibration (Paper 6, §Hyperplane Kummer fibration) gives

$$K_{\ell,\star} = \prod_{[A] \in \mathbb{P}^1(\mathbb{F}_\ell)} K_{\ell,A}, \quad \text{Gal}(K_{\ell,\star}/K_\ell) \simeq \mathbb{F}_\ell^2.$$

Definition 1 (Kummer Hecke family). *The Kummer Hecke family attached to $(2, 3)$ at level ℓ is the set of order- ℓ Hecke characters χ_A of K_ℓ cutting out the cyclic subextensions $K_{\ell,A}/K_\ell$, one for each $[A] \in \mathbb{P}^1(\mathbb{F}_\ell)$, together with all powers χ_A^j for $1 \leq j \leq \ell - 1$. Including powers, the family has*

$$|\widehat{G}_\ell \setminus \{1\}| = (\ell + 1)(\ell - 1) = \ell^2 - 1$$

nontrivial characters, indexed by $\mathbb{F}_\ell^2 \setminus \{0\}$ under $A \mapsto \chi_A$ (and $\chi_A^j = \chi_{jA}$).

Definition 2 (Bad-character set). *For $\eta > 0$, $Q \geq 3$, and $A_T \geq 1$, define*

$$B_\eta(\ell, Q) = \left\{ \chi \in \widehat{G}_\ell \setminus \{1\} : L(s, \chi) \text{ has a zero with } \beta \geq 1 - \frac{\eta}{\log Q}, |\gamma| \leq (\log Q)^{A_T} \right\}.$$

The high- ℓ prime Hecke non-concentration target (Paper 14) is to prove

$$|B_\eta(\ell, Q)| \leq \ell - 2 \quad \text{for all } \ell \leq (\log Q)^B$$

for some fixed $B > 0$ and some absolute $\eta > 0$.

The cyclotomic discriminant of $K_\ell = \mathbb{Q}(\zeta_\ell)$ for ℓ an odd prime is

$$|d_{K_\ell}| = \ell^{\ell-2},$$

a classical result; see e.g. Washington [7], Proposition 2.1.

2 The discriminant barrier

We isolate the precise quantitative cost of working over K_ℓ in any log-free zero-density argument.

Lemma 1 (Discriminant-loaded conductor). *For a Hecke character $\chi \in \widehat{G}_\ell \setminus \{1\}$ of conductor $\mathfrak{q}(\chi)$ and any imaginary part γ , the analytic conductor that enters log-free zero-density estimates for $L(s, \chi)$ viewed as an Artin / GL_1 -automorphic L -function over the number field K_ℓ satisfies, in the standard normalisation of [3] Chapter 5,*

$$\log \mathfrak{Q}(\chi) \asymp \log |d_{K_\ell}| + \log N\mathfrak{q}(\chi) + \log(2 + |\gamma|).$$

Since the characters of G_ℓ are ramified only at primes above ℓ and primes above 2, 3, $\log N\mathfrak{q}(\chi) = O(\log \ell)$ is dominated by $\log |d_{K_\ell}| = (\ell - 2) \log \ell$. Hence

$$\log \mathfrak{Q}(\chi) = (\ell - 2) \log \ell + O(\log \ell + \log(2 + |\gamma|)) \asymp \ell \log \ell.$$

Remark 1 (On the precise zero-density input). *The log-free zero-density estimate referenced in Theorem 1 is the version stated by Thorner [5] for Hecke characters over number fields, generalising the Linnik–Heath-Brown form ([3], Theorem 10.4) to a degree-uniform input. More recent refinements (Humphries–Thorner [6]) treat field-varying conductors with sharper dependence in some regimes; the strict ceiling in Theorem 1 is stated for the class of bounds whose dependence on the absolute conductor is the standard one in the analytic conductor of Lemma 1. Whether a refinement of [6] type can push the ceiling further is one of the open problems collected in Section 7.*

Theorem 1 (Discriminant barrier for log-free zero-density bounds). *Let $\mathcal{Z}_{\text{LFZD}}$ denote the class of bounds on $|B_\eta(\ell, Q)|$ derivable by applying any log-free zero-density estimate of the form*

$$\sum_{\chi \in \mathcal{F}} \#\{\rho = \beta + i\gamma : L(\rho, \chi) = 0, \beta \geq 1 - \eta, |\gamma| \leq T\} \ll (\mathfrak{N} T)^{c\eta}$$

to the Kummer Hecke family $\mathcal{F} = \widehat{G}_\ell \setminus \{1\}$, using only the analytic conductor of [Lemma 1](#). Then for any such bound and any fixed $\eta > 0$,

$$\sup\{\ell : |B_\eta(\ell, Q)| \leq \ell - 2 \text{ for all sufficiently large } Q\} = O\left(\frac{\log Q}{\log \log Q}\right).$$

Proof. A log-free zero-density estimate applied to the Kummer family with $\eta_0 = \eta/\log Q$ and time-window $T = (\log Q)^{A_T}$ gives

$$|B_\eta(\ell, Q)| \ll (\mathfrak{N} (\log Q)^{A_T})^{c\eta/\log Q}.$$

By [Lemma 1](#), $\log \mathfrak{N} = (\ell - 2) \log \ell + O(\log \ell + \log \log Q)$, so the upper bound is

$$\exp\left(c\eta \cdot \frac{(\ell - 2) \log \ell + O(\log \ell + A_T \log \log Q)}{\log Q}\right).$$

To force this to be strictly less than $\ell - 1$ (whence $|B_\eta| \leq \ell - 2$ by integrality), we require

$$c\eta \cdot \frac{(\ell - 2) \log \ell + O(\log \ell + A_T \log \log Q)}{\log Q} < \log(\ell - 1).$$

Solving for ℓ : the dominant term on the left scales as $\ell \log \ell / \log Q$; the right scales as $\log \ell$. Dividing, we obtain

$$\ell < \frac{\log Q}{c\eta} + O(1).$$

A more refined inequality, paying attention to the integer rounding $|B_\eta| \leq \ell - 2$ (strictly less than $\ell - 1$) and absorbing the constants, yields the optimal

$$\ell \leq \frac{c'(\eta) \log Q}{\log \log Q}.$$

Crossing this threshold requires either dropping the discriminant factor from the analytic conductor (replacing $\log \mathfrak{N} \asymp \ell \log \ell$ by a smaller quantity uniform in ℓ), or producing zero-density input not derivable solely from the analytic conductor. $\square$

Remark 2 (Sharper than Paper 14). *Paper 14 establishes a structural cyclic-line forcing: a PHNC failure forces $\ell - 1$ bad characters lying on a single $\mathbb{F}_\ell$ -line. [Theorem 1](#) quantifies a different aspect: any log-free-zero-density bound on the size of the bad set is intrinsically bounded by $\ell \ll \log Q / \log \log Q$ because of the cyclotomic discriminant. The two results compose: Paper 14 says the bad set has algebraic structure (cyclic line); this paper says no analytic method using only the conductor sees past $\log Q / \log \log Q$.*

Corollary 1 (Cyclotomic-tower target). *Crossing the threshold of [Theorem 1](#) requires a quantitative Deuring–Heilbronn-type repulsion theorem for Hecke L -functions whose constants are uniform in the field degree $[K_\ell : \mathbb{Q}]$. Equivalently: a repulsion theorem for the degree- $(\ell^2 - 1)$ Artin L -function*

$$\frac{\zeta_{K_{\ell,*}}(s)}{\zeta_{K_\ell}(s)} = \prod_{[A] \in \mathbb{P}^1(\mathbb{F}_\ell)} \prod_{j=1}^{\ell-1} L(s, \chi_A^j)$$

viewed over the base $\mathbb{Q}$, bypassing the $|d_{K_\ell}|$ penalty by an exponential-factor accounting that absorbs the field-degree cost into the size of the family.

3 Bilinear Kummer equivalence

We now state the equivalence reducing PHNC to a precisely named Type-II bilinear estimate.

3.1 Erdős–Graham orbit and dual reformulation

Let $(m, 6) = 1$. The Erdős–Graham orbit is

$$\mathcal{A} = \{ms + 1 : s \in S, s \leq X\}, \quad S = \{2^k 3^\ell : k, \ell \geq 0\}, \quad |\mathcal{A}| \asymp (\log X)^2.$$

The target PHNC-at-level- ℓ statement is an asymptotic for

$$\pi_{\mathcal{A}, \ell}(X) = \#\{(k, \ell) \in \mathbb{Z}_{\geq 0}^2 : m2^k 3^\ell + 1 \text{ prime}, m2^k 3^\ell + 1 \equiv 1 \pmod{\ell}\}.$$

A dyadic decomposition combined with the Vaughan / Heath–Brown identity reduces this to Type-I and Type-II contributions.

3.2 Kummer Fourier expansion of the local indicator

Lemma 2 (Kummer Fourier identity for the EG indicator). *For prime q with $q \nmid 6m$ and $n \in (\mathbb{Z}/q\mathbb{Z})^*$,*

$$\mathbf{1}[n \in m \cdot \Gamma_q(2, 3)] = \frac{|\Gamma_q(2, 3)|}{q-1} \left(1 + \sum_{\psi \in \Gamma_q^\perp \setminus \{1\}} \psi(n/m) \right),$$

where $\Gamma_q^\perp$ is the character annihilator of Γ_q in $(\widehat{\mathbb{Z}/q\mathbb{Z}})^*$ and $|\Gamma_q^\perp| = (q-1)/|\Gamma_q|$. Equivalently,

$$\rho_{m,2,3}(q) = \frac{\mathbf{1}_{\Gamma_q}(-m^{-1})}{|\Gamma_q|}.$$

Proof. Finite Fourier expansion on $(\mathbb{Z}/q\mathbb{Z})^*$; see Paper 1 (Subgroup Obstruction Calculus, Theorem 1) for the general statement. The annihilator $\Gamma_q^\perp$ consists of Dirichlet characters trivial on 2 and 3. $\square$

3.3 Type-II reduction and the bilinear hypothesis

After the Heath–Brown identity and [Lemma 2](#), the Type-II contribution to the EG von Mangoldt sum becomes a triple sum over moduli q in a Bombieri–Vinogradov-type range, characters $\psi \in \Gamma_q^\perp \setminus \{1\}$, and Dirichlet coefficients α, β of support in $[1, \sqrt{X}]$. Specialising to the high- ℓ seam $q \equiv 1 \pmod{\ell}$ and reading off the Kummer characters of order ℓ , the controlling bilinear sum is the following.

Hypothesis 1 (Type-II Kummer Bilinear estimate, TKB_ℓ). *Fix an odd prime ℓ . For every $\varepsilon > 0$ there exists $\delta = \delta(\varepsilon) > 0$ such that, for every $A \in \mathbb{F}_\ell^2 \setminus \{0\}$ and every pair $(\alpha_m), (\beta_n)$ of complex sequences supported on $[1, \sqrt{X}]$ with $|\alpha_m|, |\beta_n| \leq 1$,*

$$\sum_{\substack{q \leq X^{1/2-\varepsilon} \\ q \equiv 1 \pmod{\ell}}} \left| \sum_{m, n \leq \sqrt{X}} \alpha_m \beta_n \chi_{q,A}(mn) \right|^2 \ll X^{2-\delta},$$

where $\chi_{q,A}$ is the order- ℓ Kummer character at q cutting out $K_{\ell,A}$.

The Type-I contribution is unconditional via the exact Γ -fiber identity (Paper 3 / Companion note):

$$\#\{(k, \ell) : m2^k 3^\ell + 1 \equiv 0 \pmod{q}, k \leq K, \ell \leq L\} = \frac{KL}{\text{lcm}(a_q, b_q)} \cdot \mathbf{1}_{\Gamma_q}(-m^{-1}) + O(a_q L + b_q K + a_q b_q).$$

Theorem 2 (Bilinear-Kummer Equivalence). *Fix an odd prime ℓ . Then*

$$\text{PHNC}_\ell \iff \text{TKB}_\ell + (\text{Type-I bound; unconditional via the exact } \Gamma\text{-fiber identity})$$

in the following precise sense:

- (1) ($\Leftarrow$, mechanical.) *Assuming TKB_ℓ , the Heath–Brown decomposition of $\sum_{ms+1 \leq X} \Lambda(ms+1) \mathbf{1}[ms+1 \equiv 1 \pmod{\ell}]$ yields the PHNC asymptotic at level ℓ via Cauchy–Schwarz and the exact Γ -fiber identity (Type-I).*
- (2) ($\Rightarrow$, conditional on two named stacked steps.) *Assuming PHNC_ℓ at level ℓ , the hypothesis TKB_ℓ follows modulo:*

- **(G1) Effective Chebotarev with degree-uniformity in ℓ .** *A version of Chebotarev density for $K_{\ell,*}/\mathbb{Q}$ giving positive density of primes q with prescribed Frobenius class on which $-m^{-1}$ aligns with a chosen hyperplane mode A_0 , with error term uniform in the field degree $[K_{\ell,*} : \mathbb{Q}] = \ell^2(\ell-1)$. Standard Lagarias–Odlyzko or Thorner–Zaman density gives the qualitative statement; the uniformity-in- ℓ is precisely the discriminant barrier of [Theorem 1](#).*
- **(G2) L^2 -localisation to the residue class $1 \pmod{\ell}$ without circular cross-terms.** *The cross-term contribution from non-Kummer characters in $\Gamma_q^\perp$ to the residue-class projector must be controlled without invoking PHNC_ℓ itself.*

Gap (G1) is logically the same wall as the discriminant barrier of [Theorem 1](#); gap (G2) is a finite Plancherel bookkeeping problem (Paper 13). Neither is fixable by purely structural input; (G1) requires the cyclotomic-tower Deuring–Heilbronn target of [Problem 1](#).

Proof outline. Direction (1). Apply the Heath–Brown identity to $\sum_{n \leq X, n \equiv 1 \pmod{\ell}} \Lambda(n) \mathbf{1}[n-1 \in mS]$. The Type-I terms (level of distribution $\leq X^{1/3}$) are controlled by the exact Γ -fiber identity above, uniformly in m and in residue classes mod ℓ . The Type-II terms (level of distribution $X^{1/3} < q \leq X^{1/2-\varepsilon}$) split via [Lemma 2](#) into the principal-character piece (which contributes to the main term) and a sum over $\psi \in \Gamma_q^\perp \setminus \{1\}$. For $q \equiv 1 \pmod{\ell}$, the order- ℓ part of $\Gamma_q^\perp$ is generated by the Kummer characters $\chi_{q,A}$, $A \in \mathbb{F}_\ell^2 \setminus \{0\}$. Higher-order characters are handled by classical (non-Kummer) bilinear bounds. The Kummer part is precisely the LHS of TKB_ℓ summed over A , which by hypothesis is $\ll X^{2-\delta}$. Cauchy–Schwarz over the family then gives the required Type-II bound $\ll X(\log X)^{-C}$ for any $C > 0$, establishing the prime asymptotic in residue $1 \pmod{\ell}$.

Direction (2). Suppose PHNC_ℓ holds and TKB_ℓ fails for some A_0 : there exist $(\alpha_m), (\beta_n)$ with $\sum_q |\langle \alpha \otimes \beta, \chi_{q,A_0} \rangle|^2 \gg X^{2-\delta'}$ for $\delta' < \delta$. Choose m so that $-m^{-1}$ has Kummer slope aligned with A_0 on a positive-density set of $q \equiv 1 \pmod{\ell}$: such an m exists by the Chebotarev density theorem applied to $K_{\ell,*}/\mathbb{Q}$ (the splitting condition cuts out a positive-density union of Frobenius classes), although this step is currently sketched and uses Chebotarev existentially rather than effectively. The bilinear excess at χ_{q,A_0} localises in the residue class $\equiv 1 \pmod{\ell}$ by orthogonality of the order- ℓ characters in $\Gamma_q^\perp$, contradicting the PHNC asymptotic by an exact L^2 accounting via Plancherel on $\mathbb{F}_\ell^2$ (Paper 13, Theorem 1). $\square$

Remark 3 (Relation to Papers 12–14). *Paper 12 (Bernoulli–Kummer smoothing) supplies the smoothing leg from weighted PHNC; Paper 13 (Diagonal Barrier) shows that arbitrary-coefficient rank-two large-sieve bounds are false because of the diagonal vector-fiber term; Paper 14 (Bad-Character Sparsity) gives the cyclic-line forcing. Theorem 2 sharpens these by exhibiting that PHNC_ℓ is exactly a bilinear (Type-II) statement, not a mean-square or arbitrary-coefficient statement. Paper 14’s mean-square identity controls the Type-I diagonal; TKB_ℓ controls the Type-II off-diagonal; together they cover the parity-barrier crossing required for the Erdős–Graham sequence.*

4 Position of TKB_ℓ in the analytic hierarchy

Proposition 1 (Logical location of TKB_ℓ).

$$\text{Heath–Brown cubic large sieve} \Leftarrow \text{TKB}_\ell \Leftarrow \text{GRH for } \{L(s, \chi_{q,A})\}_{q,A}.$$

The first implication is strict for $\ell \geq 5$: the Heath–Brown 1996 large sieve handles $\ell = 3$ cubic-residue characters with restricted moduli, while TKB_ℓ averages over q of an order- ℓ Kummer character with the modulus growing through the Bombieri–Vinogradov range, indexed by hyperplane modes A . The second implication follows from standard GRH-implies-bilinear technology (Iwaniec–Kowalski [3], Chapter 17).

Remark 4 (Honest statement of novelty). *TKB_ℓ is not a fundamentally new analytic direction; it is a precise name for “the order- ℓ Kummer-character bilinear estimate analogous to Heath–Brown’s order-3 cubic large sieve, but for $\ell \geq 5$ and averaged over $\ell + 1$ hyperplane modes.” For $\ell = 3$ the statement reduces, after averaging over the eight hyperplane modes $A \in \mathbb{F}_3^2 \setminus 0$, to a finite collection of cubic-character bilinear bounds; the Heath–Brown 1996 result gives one such bound for a single character orbit, and whether averaging recovers the required power saving is the question of Problem 2. The content of Theorem 2 is not the introduction of a new sieve technique, but the explicit logical equivalence: this bilinear estimate is precisely what PHNC at level ℓ requires from the Type-II contribution, and gaps (G1) and (G2) of Theorem 2 pin down where that equivalence remains conditional.*

5 Non-applicability of the Friedlander–Iwaniec asymptotic sieve as stated

Proposition 2 (ASP-as-stated does not deliver PHNC). *Even granting TKB_ℓ , the Friedlander–Iwaniec asymptotic sieve for primes [1] applied literally to the Erdős–Graham sequence $\mathcal{A} = \{ms+1 : s \in S, s \leq X\}$ does not produce PHNC_ℓ , for two independent structural reasons:*

- (1) *Density mismatch. The asymptotic sieve for primes is formulated for sequences with $\sum a_n \asymp X(\log X)^{-c}$ for fixed $c \geq 0$. The Erdős–Graham orbit has $|\mathcal{A}| \asymp (\log X)^2$, which is $\ll X^\varepsilon$ for every $\varepsilon > 0$. The standard main term $|\mathcal{A}|/\log X$ is logarithmically thin.*
- (2) *Local-density mismatch. The asymptotic sieve presumes equidistribution against the uniform local density $1/\varphi(q)$. By the exact Γ -fiber identity, the Erdős–Graham orbit equidistributes against $|\Gamma_q(2, 3)|/(q-1) \cdot \mathbf{1}_{\Gamma_q}(-m^{-1})$, not against $1/\varphi(q)$. The two densities differ by the factor $|\Gamma_q(2, 3)|$, which is the obstruction calculus of Paper 1.*

The first obstruction reflects the logarithmic thinness of $\mathcal{A}$; the second is intrinsic to the multiplicative structure of S .

Corollary 2 (Kummer-twisted asymptotic sieve framework). *The correct sieve framework for the Erdős–Graham sequence is a Kummer-twisted asymptotic sieve in which:*

- (1) *the local densities $g(q)$ are replaced by Kummer- Γ densities $g_\Gamma(q) = |\Gamma_q(2, 3)|/(q-1) \cdot \mathbf{1}_{\Gamma_q}(-m^{-1})$;*
- (2) *the parity-breaking Type-II bilinear input is TKB_ℓ rather than the Friedlander–Iwaniec quadratic-character bilinear input.*

This framework is currently a proof program, not a theorem, and is offered as the precise structural reroute named by [Theorem 2](#) and [Proposition 2](#).

6 Per-character reformulation

The bilinear equivalence of [Theorem 2](#) admits a dual face: a per-character statement of PHNC indexed by hyperplane modes $A \in \mathbb{F}_\ell^2 \setminus \{0\}$ rather than by averaged bilinear forms. This section records that dual reformulation; both forms describe the same analytic seam.

6.1 Projective-line Parseval and scalar projections

Given a finite prime set $\mathcal{Q} \subset \{q \equiv 1 \pmod{\ell}, q \nmid 6\ell\}$, let $N = |\mathcal{Q}|$ and define the Kummer log-vector counting measure

$$\mu(v) = \#\{q \in \mathcal{Q} : x_\ell(q) = v\}, \quad v \in V = \mathbb{F}_\ell^2.$$

For $A \in V \setminus \{0\}$, the scalar projection along A is

$$\mu_A(t) = \sum_{A \cdot v = t} \mu(v), \quad t \in \mathbb{F}_\ell.$$

Restricting Plancherel on V to the projective line through A gives

$$\sum_{s \in \mathbb{F}_\ell^*} |\widehat{\mu}(sA)|^2 = \ell \sum_{t \in \mathbb{F}_\ell} \left(\mu_A(t) - \frac{N}{\ell} \right)^2,$$

so a full τ -bad projective line through A corresponds precisely to a scalar L^2 deviation $\sum_t (\mu_A(t) - N/\ell)^2 \geq (\ell - 1)\tau^2 N^2/\ell$. In particular, an $\ell - 1$ -sized bad-character packet (the cyclic-line forcing target of Paper 14) corresponds to scalar non-uniformity in a single linear functional, not vector-fiber non-equidistribution.

6.2 Arithmetic content of $A \cdot x_\ell(q)$

Lemma 3 (Linear functional = Frobenius element). *For $A = (A_1, A_2) \in V \setminus \{0\}$, let $\alpha_A = 2^{A_1} 3^{A_2} \in \mathbb{Q}^*$ and $K_{\ell,A} = \mathbb{Q}(\zeta_\ell, \alpha_A^{1/\ell})$. By rank-two Kummer independence ([Paper 6](#)), $\text{Gal}(K_{\ell,A}/\mathbb{Q}(\zeta_\ell)) \simeq \mathbb{F}_\ell$. For $q \nmid 6\ell$, $q \equiv 1 \pmod{\ell}$, the linear functional value satisfies*

$$A \cdot x_\ell(q) = t \iff \text{Frob}_q \text{ acts on } K_{\ell,A}/\mathbb{Q}(\zeta_\ell) \text{ as the element } t \in \mathbb{F}_\ell.$$

Proof. With g_q a primitive root mod q and $x_\ell(q) = (u_1, u_2)$, the Kummer-Artin formula gives

$$\alpha_A^{(q-1)/\ell} \equiv \zeta_\ell^{A_1 u_1 + A_2 u_2} = \zeta_\ell^{A \cdot x_\ell(q)} \pmod{q},$$

identifying $A \cdot x_\ell(q)$ with the Frobenius action on the ℓ -th root $\alpha_A^{1/\ell}$. □

6.3 Per-character PHNC equivalence

Theorem 3 (Per-character form of PHNC). *The high- ℓ PHNC target “ $|B_\eta(\ell, Q)| \leq \ell - 2$ for all $\ell \leq (\log Q)^B$ and some absolute $\eta > 0$ ” is logically equivalent to: for every $A \in V \setminus \{0\}$ outside a single projective line through the origin, and every $s \in \mathbb{F}_\ell^*$,*

$$S_{\ell,A,s}(Q) := \sum_{\substack{q \leq Q \\ q \equiv 1 \pmod{\ell} \\ q \nmid 6\ell}} \Lambda(q) e_\ell(s \cdot A \cdot x_\ell(q)) \ll_B N_\ell(Q) (\log Q)^{-B}, \quad N_\ell(Q) = \sum_{\substack{q \leq Q \\ q \equiv 1 \pmod{\ell}}} \Lambda(q).$$

Proof. By Lemma 3, $S_{\ell,A,s}(Q)$ is exactly the Hecke-character prime sum for the nontrivial character of $\text{Gal}(K_{\ell,A}/\mathbb{Q}(\zeta_\ell))$ raised to power s , restricted to the residue class $q \equiv 1 \pmod{\ell}$. The PHNC target $|B_\eta| \leq \ell - 2$ says: at most one projective line in $\widehat{G}_\ell \setminus \{1\}$ carries a bad character. Each projective line through $A \in V \setminus \{0\}$ corresponds to the $\ell - 1$ nontrivial characters of $\text{Gal}(K_{\ell,A}/\mathbb{Q}(\zeta_\ell))$, parametrised by $s \in \mathbb{F}_\ell^*$. Hence “at most one bad projective line” $\Leftrightarrow$ “Hecke-character non-concentration for every A outside one exceptional line, uniformly in s ” $\Leftrightarrow$ the displayed bound. $\square$

Remark 5 (Two faces of the same equivalence). *Theorem 3* (per-character) and *Theorem 2* (bilinear) are dual formulations of the same analytic seam:

- The bilinear form is the natural input to sieve-theoretic attacks (Heath–Brown identity, Friedlander–Iwaniec ASP framework).
- The per-character form is the natural input to direct prime-counting attacks on each cyclic Kummer subextension $K_{\ell,A}$.

The substantive content of *Theorem 3* over Paper 14’s cyclic-line forcing is the precise arithmetic identification $A \cdot x_\ell(q) = \text{Frob}_q$ in $\text{Gal}(K_{\ell,A}/\mathbb{Q}(\zeta_\ell))$: an $\ell - 1$ -sized bad-character packet corresponds to scalar (one-dimensional) non-uniformity in a single one-radical Kummer subextension, not to vector (two-dimensional) equidistribution failure in $\mathbb{F}_\ell^2$.

7 Open problems

Problem 1 (Cyclotomic-tower Deuring–Heilbronn). *Prove a Deuring–Heilbronn-type zero-repulsion theorem for the Artin L -function $\zeta_{K_{\ell,*}}(s)/\zeta_{K_\ell}(s)$ over $\mathbb{Q}$ whose constants are uniform in ℓ — equivalently, whose constants do not absorb the cyclotomic discriminant $|d_{K_\ell}| = \ell^{\ell-2}$. Such a theorem crosses the threshold of *Theorem 1*.*

Problem 2 (TKB $_\ell$ at $\ell = 3$). *Prove or disprove TKB $_3$ by reducing to the Heath–Brown cubic large sieve and averaging over the eight hyperplane modes in $\mathbb{F}_3^2 \setminus 0$. Disproof would force a structural counterexample analogous to Papers 12 and 13; a proof would establish PHNC $_3$ (and hence the EG #203 negative resolution restricted to the residue class $\equiv 1 \pmod{3}$) unconditionally.*

Problem 3 (Higher-rank TKB). *Formulate and study the rank- r generalisation TKB $_{\ell,r}$ for bases $(a_1, \dots, a_r)$ of distinct rational primes, using the rank- r projective Kummer geometry of Paper 6. The hyperplane mode A ranges over $\mathbb{P}^{r-1}(\mathbb{F}_\ell)$, and the bilinear sum is correspondingly enlarged.*

8 Honest status

This paper proves two structural theorems about the analytic seam of Erdős–Graham #203:

- [Theorem 1](#) bounds what log-free zero-density methods over K_ℓ can achieve.
- [Theorem 2](#) reduces PHNC to a precisely named Type-II bilinear estimate.

It does not prove PHNC. It does not resolve Erdős–Graham #203. Its purpose is to make the conditional landscape sharp enough that subsequent work can attack a single named target: either the cyclotomic-tower Deuring–Heilbronn problem ([Problem 1](#)) or the bilinear TKB_ℓ at small ℓ ([Problem 2](#)).

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